

Deep Learning

Deep Learning is a subset of machine learning that utilizes neural networks with many layers to model complex patterns in data.

It has revolutionized fields such as computer vision, natural language processing, and speech recognition. By training on large datasets, deep learning algorithms can automatically detect features and improve their performance over time.





Understanding Neural Networks

1

ANN

General-purpose learning network

2

CNN

Feature extractor for images

3

Autoencoder

Compress then reconstruct

4

Decoder

Rebuild or generate output from learned features

Artificial Neural Network Overview

Inspired by Human Brain

Consisting of interconnected neurons (nodes).

Applications

Applied to classification, regression, and pattern recognition tasks.

Optimization Algorithms

Uses Gradient Descent, Adam.



Layer Organization

Organized into input layer, hidden layers, and output layer.

Neural Computation

Each neuron computes a weighted sum followed by an activation function.

Common Activations

Includes ReLU, Sigmoid, Tanh, Softmax.

Learning Process

Network learns by backpropagation, adjusting weights to minimize loss.



Understanding Convolutional Neural Networks

Grid-like Data

Specialized neural network designed for images.

Convolution Layers

Uses convolution layers to extract spatial features.

Pooling Layers

Reduces dimensionality and computation.

Spatial Relationships

Maintains spatial relationships between pixels.

Fully Connected Layers

Performs final classification or regression.

Automatic Feature Learning

Learns features automatically from data.

Applications

Widely used in various fields.

Autoencoders



Unsupervised Neural Network

An unsupervised neural network used for representation learning.



Encoder and Decoder

Consists of two parts: Encoder and Decoder.



Low-Dimensional Latent Space

Encoder compresses input into a low-dimensional latent space.



Reconstruction Process

Decoder reconstructs the original input from the latent representation.



Training Objective

Trained by minimizing reconstruction error.



Applications

Used for dimensionality reduction, denoising, anomaly detection



Variants

Variants include Denoising Autoencoders, Sparse Autoencoders, VAEs

Decoder in Neural Networks

Mapping

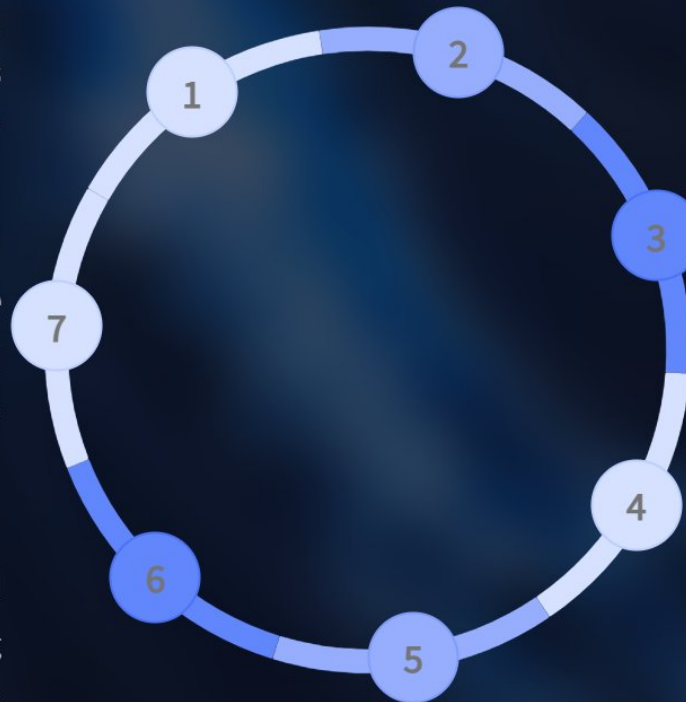
Maps latent or encoded representations back to output space.

Key Role

Plays a key role in data reconstruction and generation tasks.

Training Method

Often trained jointly with an encoder using end-to-end learning.



Applications

Commonly used in autoencoders, transformers, and sequence models.

Input/Output

Takes compressed information as input and generates structured output.

Layer Types

Uses dense, convolutional, or attention-based layers.

NLP Function

In NLP, decoders generate output token by token.